AF3625
Engineering Economics

Topic 5
International Trade and Globalization

Contents of Topic 5

- How does free international trade benefit all the countries involved?
- Why do governments intervene in trade by implementing protectionist policies?
- What are the welfare impacts of trade intervention?

Something to Think About

- Why do governments favor one group over another in their intervention in trade and globalization?
- How does the Hong Kong government protect local domestic helpers by adopting Minimum Allowable Wage (MAW) for foreign domestic helpers (see Topic 2)?

5.1 HOW GLOBAL MARKETS WORK

Imports are the goods and services people in one country buy from people in other countries.

Exports are the goods and services people in one country sell to people in other countries.

Using the US as an example and assuming the small country case (i.e. all buyers and sellers in the US are price takers), we explain why a country imports one good and exports another.

Refer to 5.1 in Self Study Notes Unit 5 if you want to know more about "comparative advantage" that will be mentioned in the example. But it is not required in this subject.

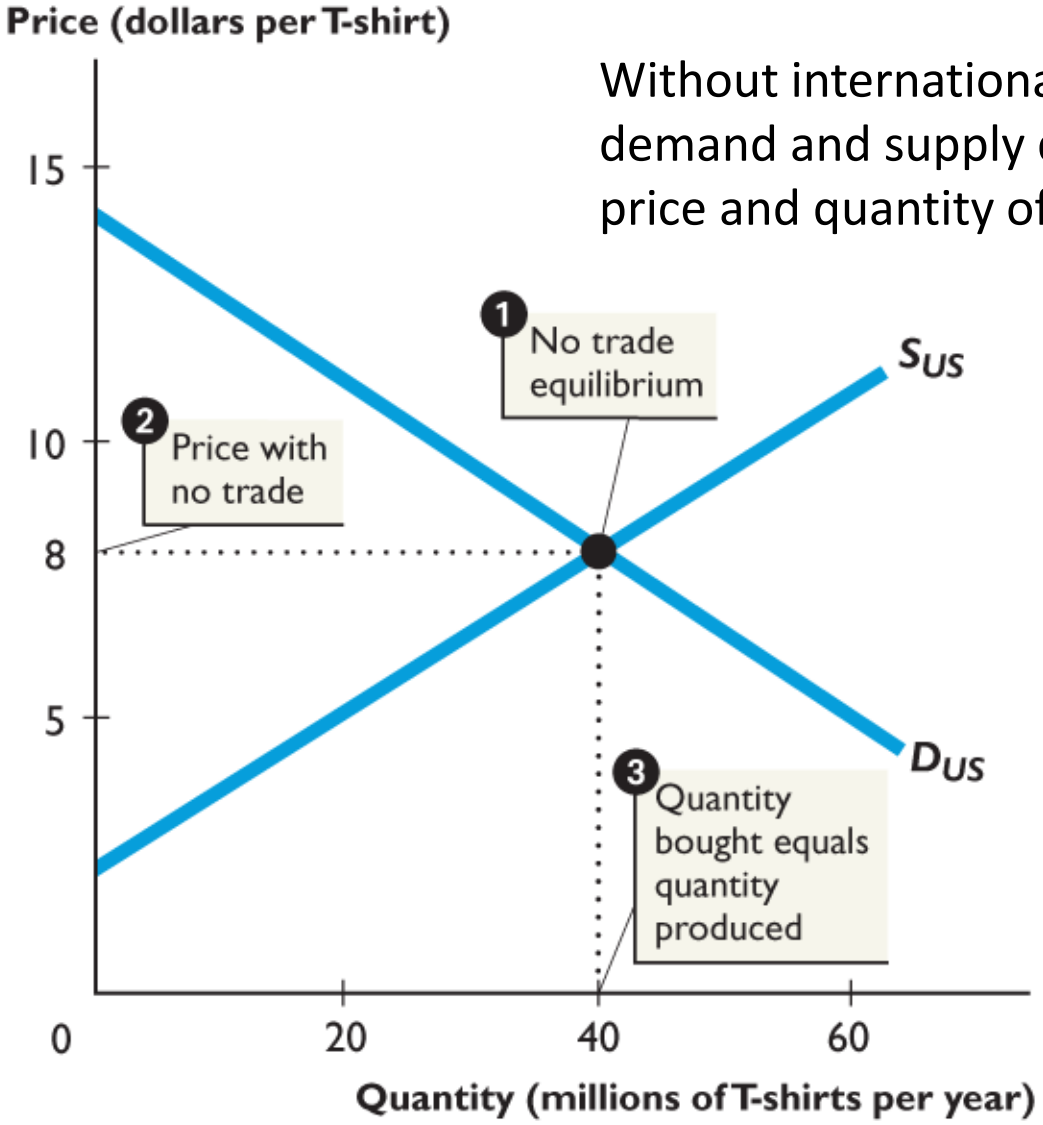


Image courtesy: aibats.com

International Trade and Social Surplus

Why does the US Import T-Shirts?

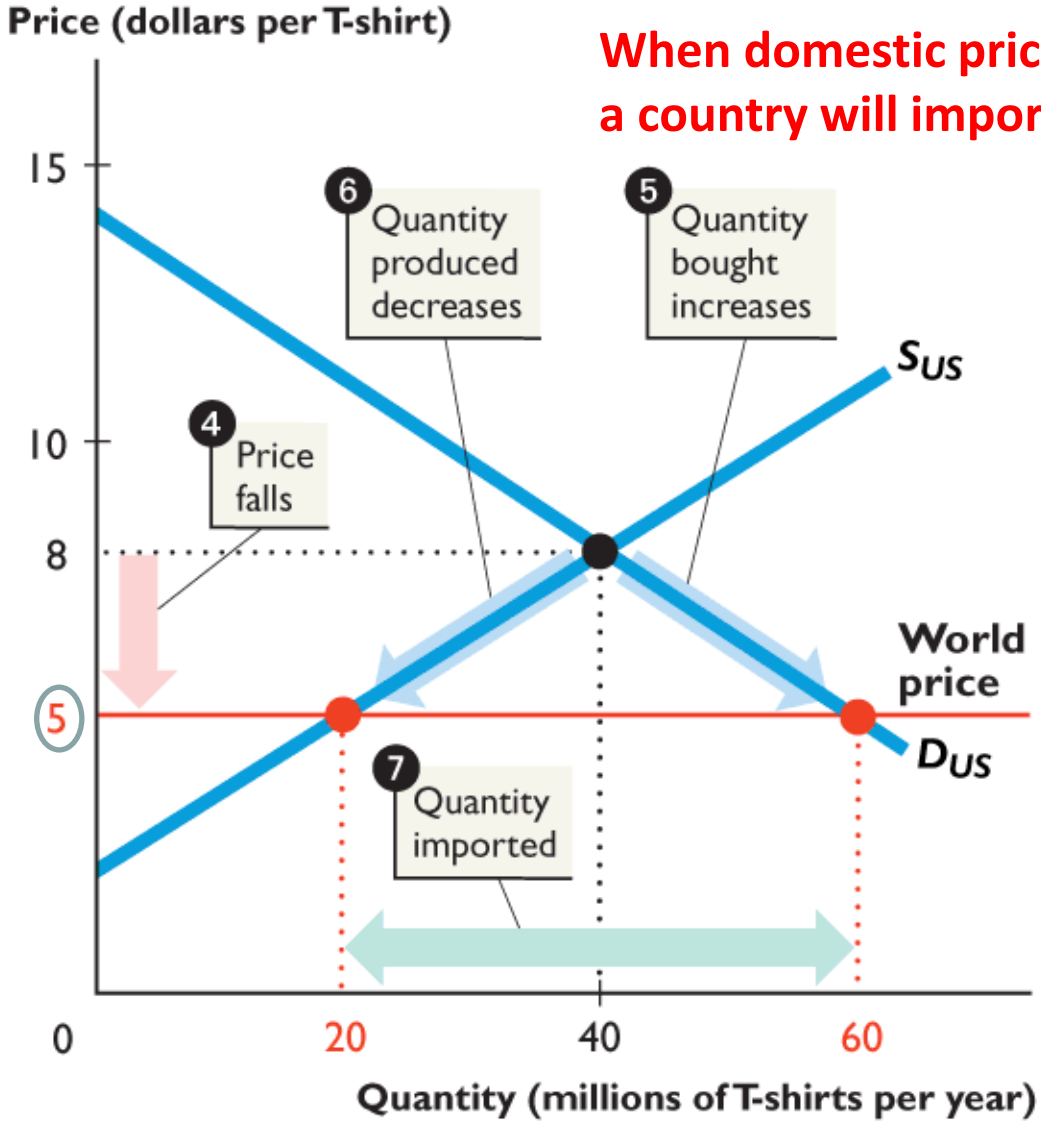
Without international trade, domestic demand and supply determine the US price and quantity of T-shirts.



(a) Equilibrium with no international trade

International Trade and Social Surplus

Why does the US Import T-Shirts?



(b) Equilibrium in a market with imports

When domestic price > world price, a country will import.

⇒ **US imports T-shirts.**

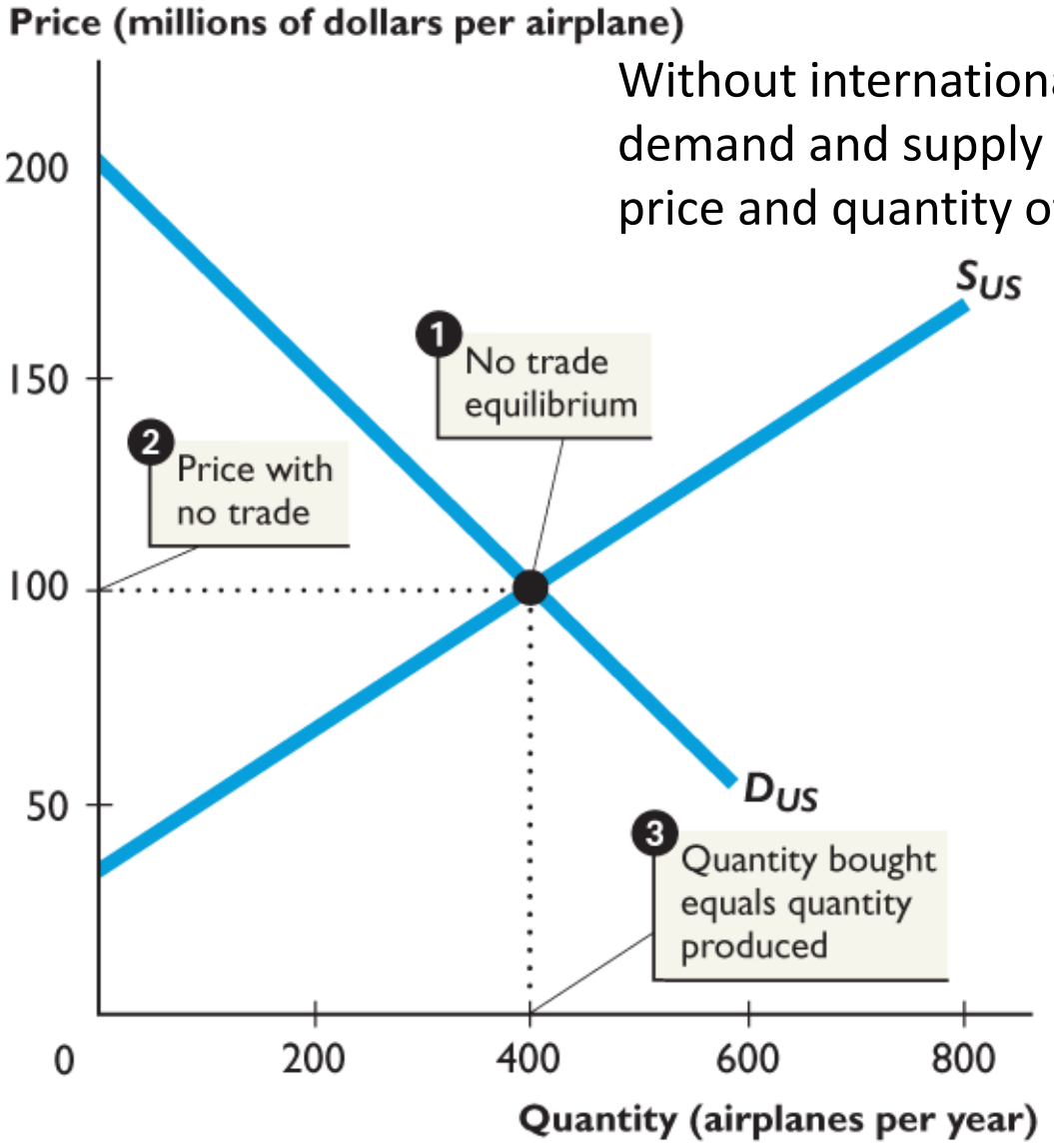
When the US opens up to trade, its price of T-shirts falls.

Qd > Qs

Q_{IM} = Qd – Qs

International Trade and Social Surplus

Why does the US export airplanes?

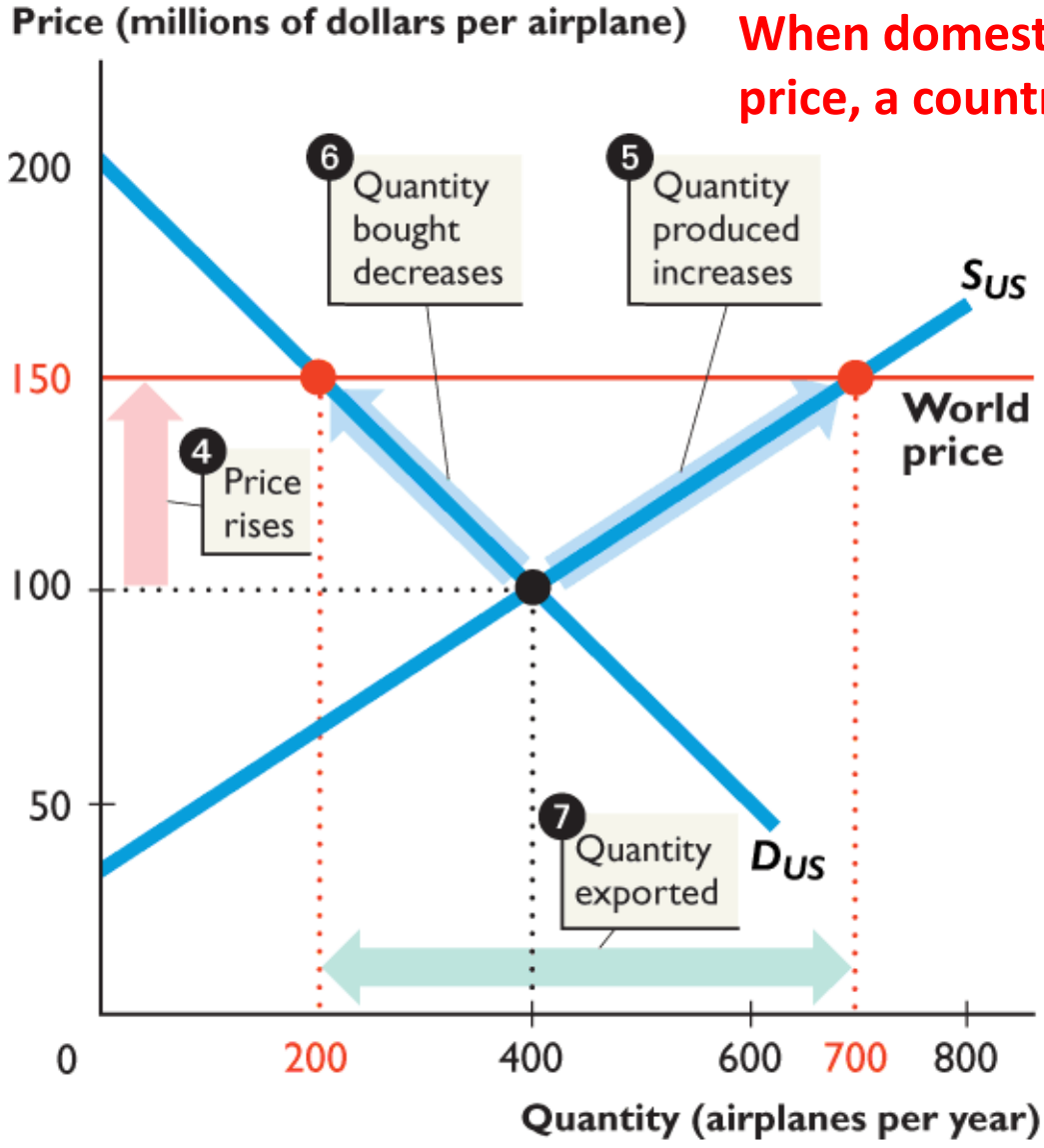


(a) Equilibrium without international trade

Without international trade, domestic demand and supply determine the US price and quantity of airplanes.

International Trade and Social Surplus

Why does the US export airplanes?



(b) Equilibrium in a market with exports

When domestic price < world price, a country will export.

⇒ US exports airplanes.

Qs > Qd

Q_{EX} = Qs – Qd

When the US opens up to trade, its price of airplanes rises.

5.2 WINNERS, LOSERS, AND NET GAINS FROM TRADE

Fill in the blanks with the appropriate word listed below.

Buyers **Sellers** **raises** **lowers**

With international trade, the import of a good _____ its domestic price while the export of a good _____ its domestic price.

_____ of imported goods benefit from lower prices and _____ of exported goods benefit from higher prices.

But not everybody gains from international competition.

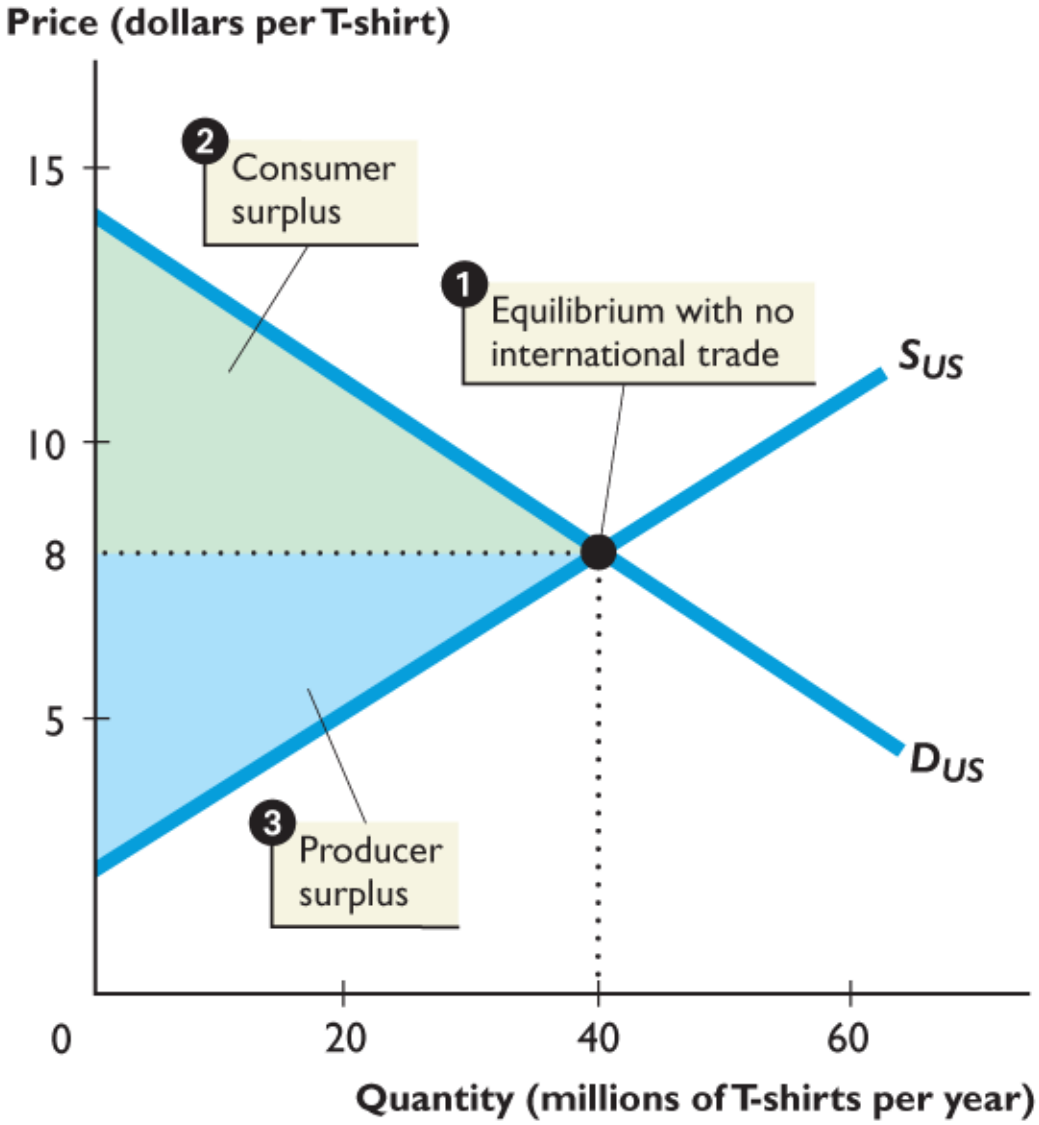
Who ***wins*** and who ***loses*** from free international trade?

We measure the gains and losses by examining the effects of international trade on consumer surplus, producer surplus, and social surplus.

International Trade and Social Surplus

The Case of Imports

Without trade, the price of a T-shirt in the US is $8 and the quantity transacted is 40 millions a year.



(a) Consumer surplus and producer surplus with no international trade

International Trade and Social Surplus

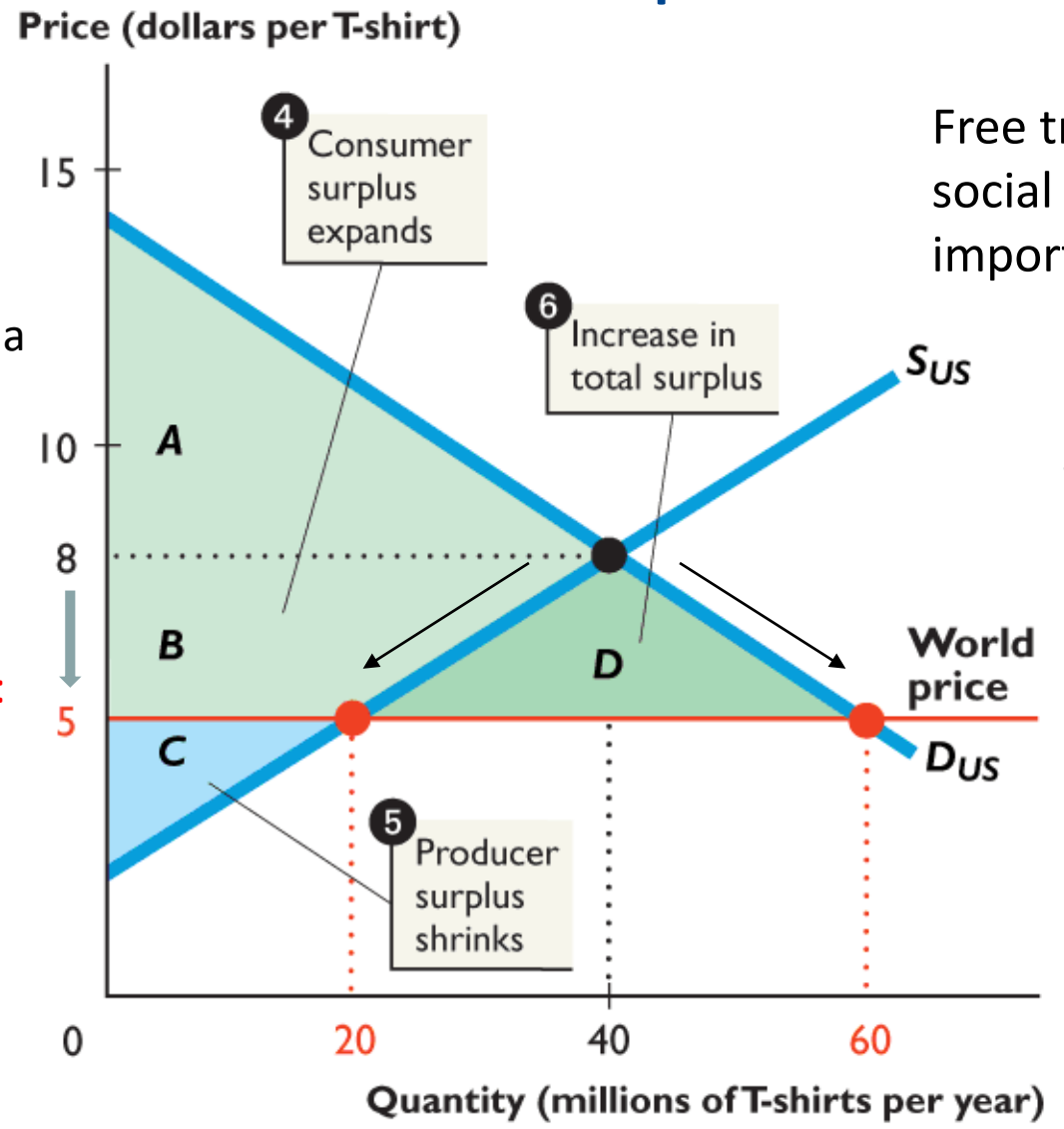
The Case of Imports

With international trade, the price of a T-shirt falls to the world price of $5.

Denoting the areas with letters:
Consumer surplus:
A → A + B + D

Producer surplus:
B + C → C

Net gain: D



(b) Gains and losses from imports

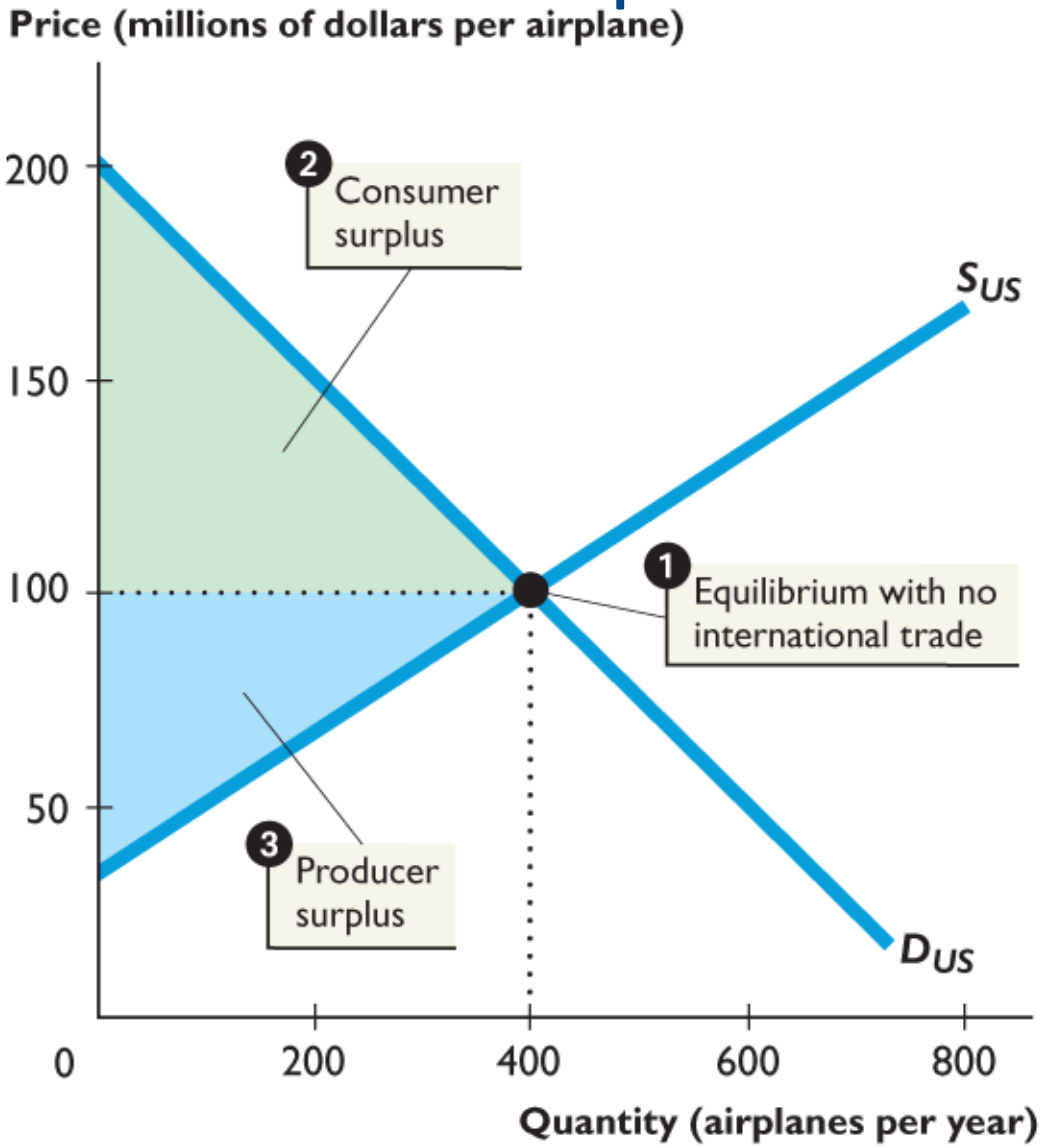
Free trade increases the social surplus of the importing country.

However, there are **winners** (consumers) and **losers** (producers).

International Trade and Social Surplus

The Case of Exports

Without trade, the price of an airplane in the US is $100 million and the quantity transacted is 400 a year.



(a) Consumer surplus and producer surplus with no international trade

International Trade and Social Surplus

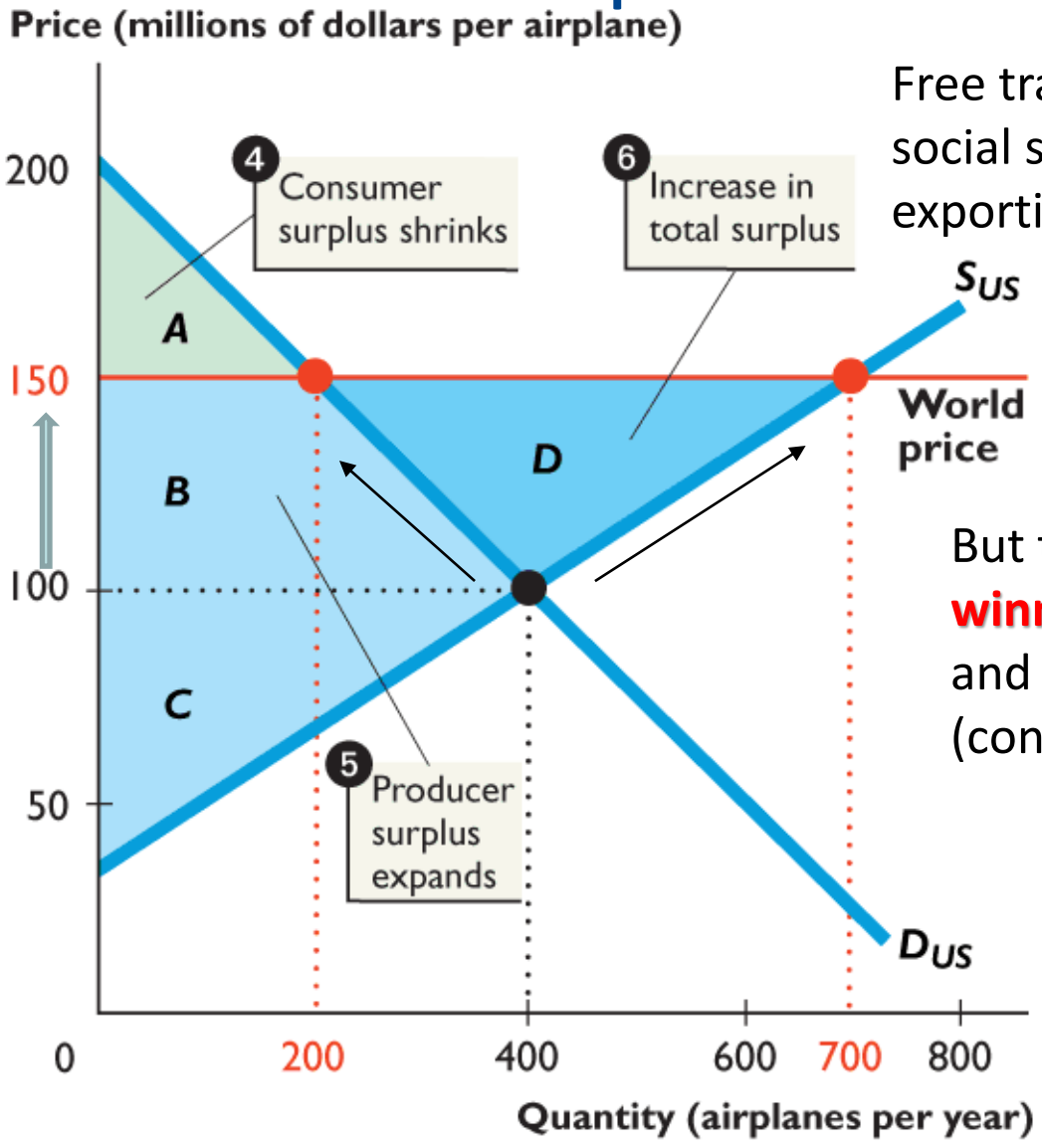
The Case of Exports

With trade, the price of an airplane rises to the world price of $150 million.

Denoting the areas with letters:
Consumer surplus:
A + B → A

Producer surplus:
C → B + C + D

Net gain: D



(b) Gains and losses from exports

Free trade increases the social surplus of the exporting country.

But there are also **winners** (producers) and **losers** (consumers).

SELF-ASSESSMENT

Fill in the blanks with the appropriate word(s) listed below.

domestic market **world market** **decreases/falls** **increases/rises** **export** **import**

When domestic price of T-shirts in the US is higher than the world price, there will be incentives for US buyers to buy the T-shirts in the _____. The US will then _____ T-shirts. After trade, US domestic price _____, producer surplus _____, and consumer surplus _____.

When the world price of airplanes is higher than domestic price in the US, there will be incentives for US producers to sell the airplanes in the _____. The US will then _____ airplanes. After trade, domestic price _____, producer surplus _____, and consumer surplus _____.

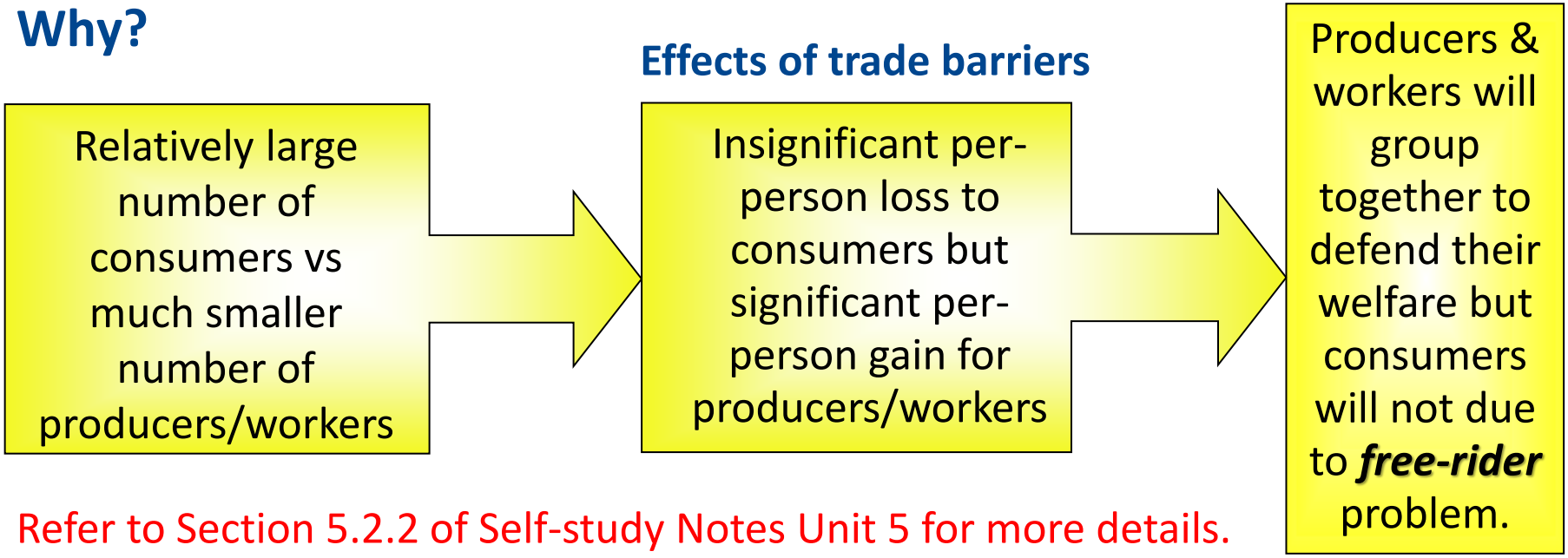
While international trade is beneficial to a country as a whole, it need not benefit everybody in the country.

SOMETHING TO THINK ABOUT

Why do governments favor one group over another in their intervention in trade and globalization?

When producers and workers are the losers, they lobby the government to erect trade barriers to protect their welfare.
When consumers are the losers, they tend not to do anything!

Why?



Refer to Section 5.2.2 of Self-study Notes Unit 5 for more details.

5.3 GOVERNMENT INTERVENTION IN INTERNATIONAL TRADE

Governments usually intervene in international trade to protect domestic producers from foreign competition.

The interventions have the following intended effects:

- Making imported goods less competitive or less available (e.g. tariffs and quotas)
- Making domestic goods more competitive in the world market (e.g. export subsidies)
- Creating trouble for foreign firms to sell in domestic market (e.g. local content requirements and administrative policies)

We will only analyse the effects of tariffs and quotas in this subject.

See Section 5.2.3 in Self-study Notes for more detailed discussion of these two types of trade barriers.

5.3 GOVERNMENT INTERVENTION IN INTERNATIONAL TRADE

Tariffs: A tariff is a tax on an imported good.

Hong Kong is a free port. There is no customs tariff on goods imported into Hong Kong.
The HKSAR Government collects an excise duty (tax) on only four types of goods irrespective of whether they are imported or locally manufactured.











The effects of an excise tax have been introduced in Topic 2.

See this link if you want to know more about the types and duty rates in Hong Kong:
http://www.customs.gov.hk/en/trade_facilitation/dutiable/types/

The Effects of a Tariff

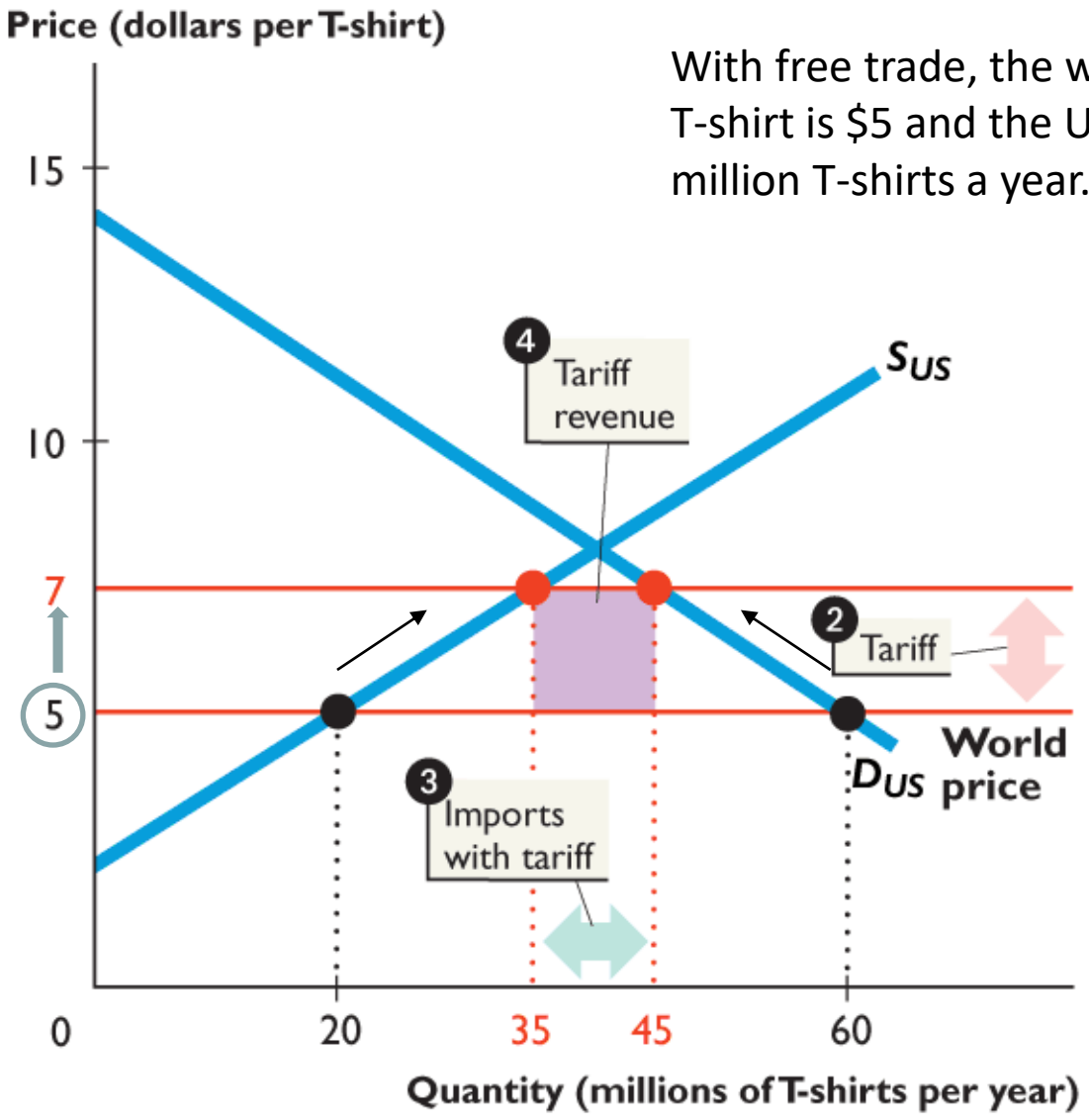
With free trade, the world price of a T-shirt is $5 and the US imports 40 million T-shirts a year.

The US imposes a tariff of $2 on each T-shirt imported.

⇒ The US price of a T-shirts rises by $2.

⇒ US imports decrease to 10 millions.

⇒ US government collects $20 million of tariff.



(b) Market with tariff

Winners, Losers, and Social Loss from a Tariff

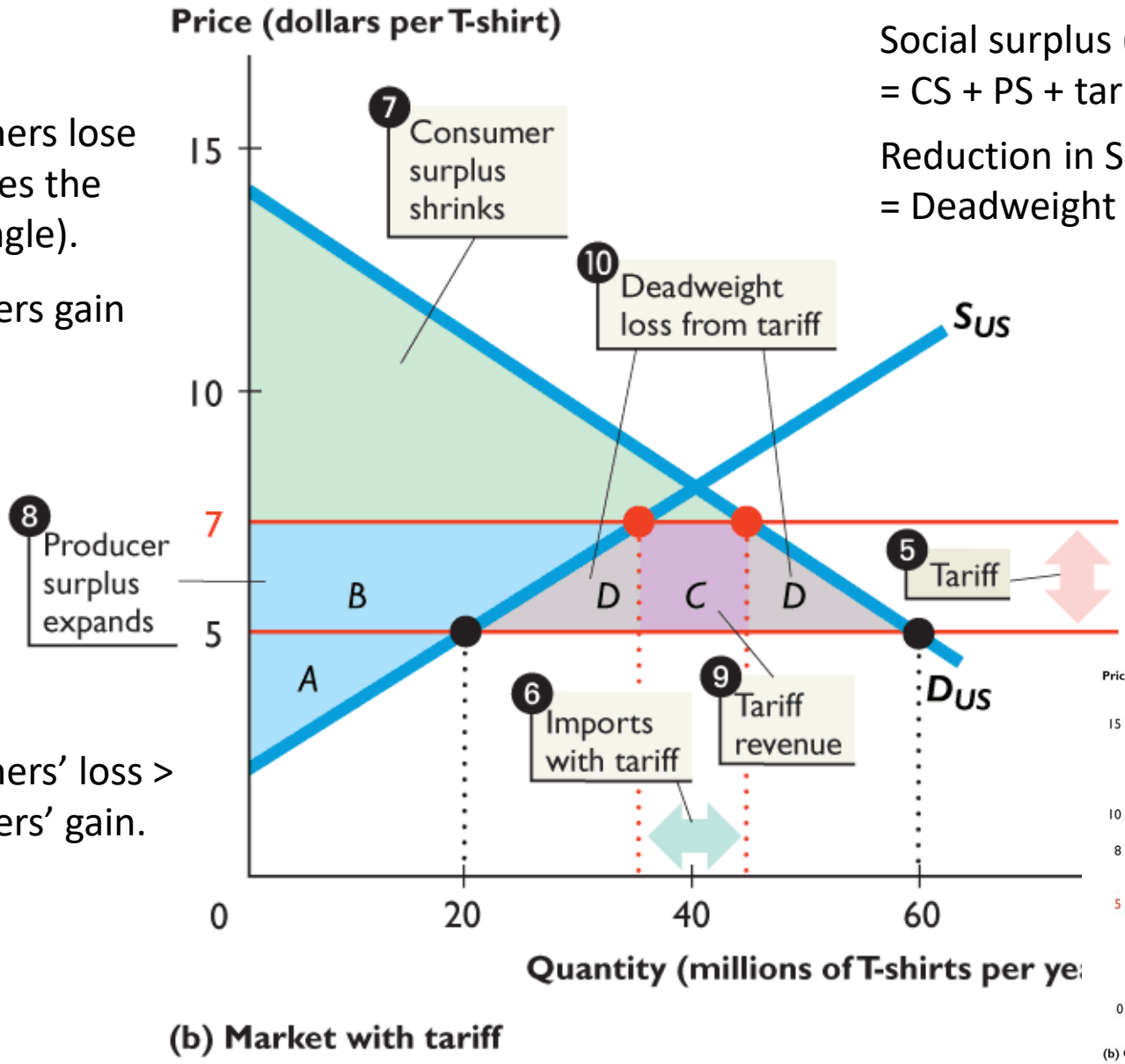
US consumers lose (CS becomes the green triangle).

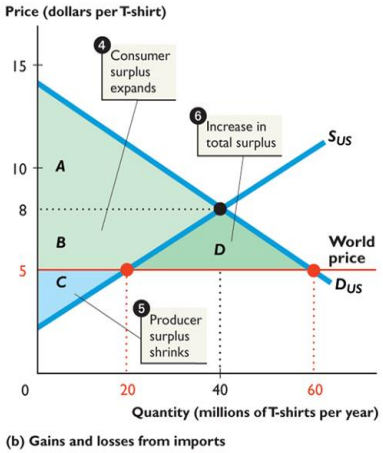
US producers gain (PS = A+B).

US consumers' loss > US producers' gain.

Social surplus (SS)
= CS + PS + tariff revenue (C)

Reduction in SS due to tariff
= Deadweight loss (DWL) = D+D





5.3 GOVERNMENT INTERVENTION IN INTERNATIONAL TRADE

Import Quotas

An **import quota** is a restriction that limits the maximum quantity of a good that may be imported in a given period.

Voluntary Export Restraint (VER)

VER has similar effects as a quota but is imposed by the exporting country at the request or suggestion of the government of the importing country.

A 1981 voluntary restraint agreement limited the Japanese to exporting cars to the U.S. annually.

The Effects of an Import Quota

The US imposes an import quota of **10 million T-shirts a year**.

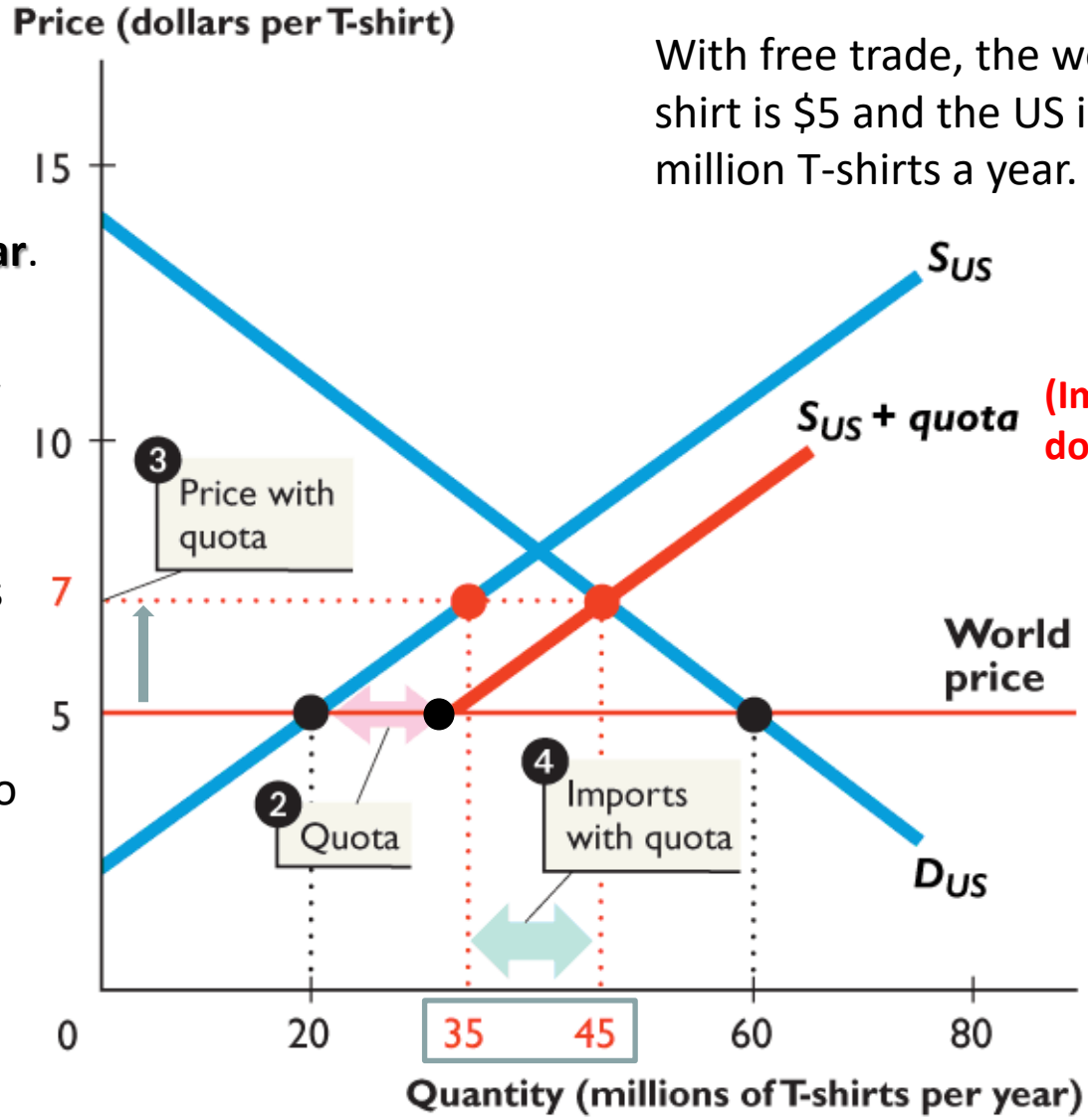
⇒ When domestic price ≥ $5, supply in US becomes S_{US} + quota

⇒ The US price rises to $7.

⇒ US producers increase output to 35 millions.

⇒ Imports fall from 40 to 10 millions.

With free trade, the world price of a T-shirt is $5 and the US imports 40 million T-shirts a year.



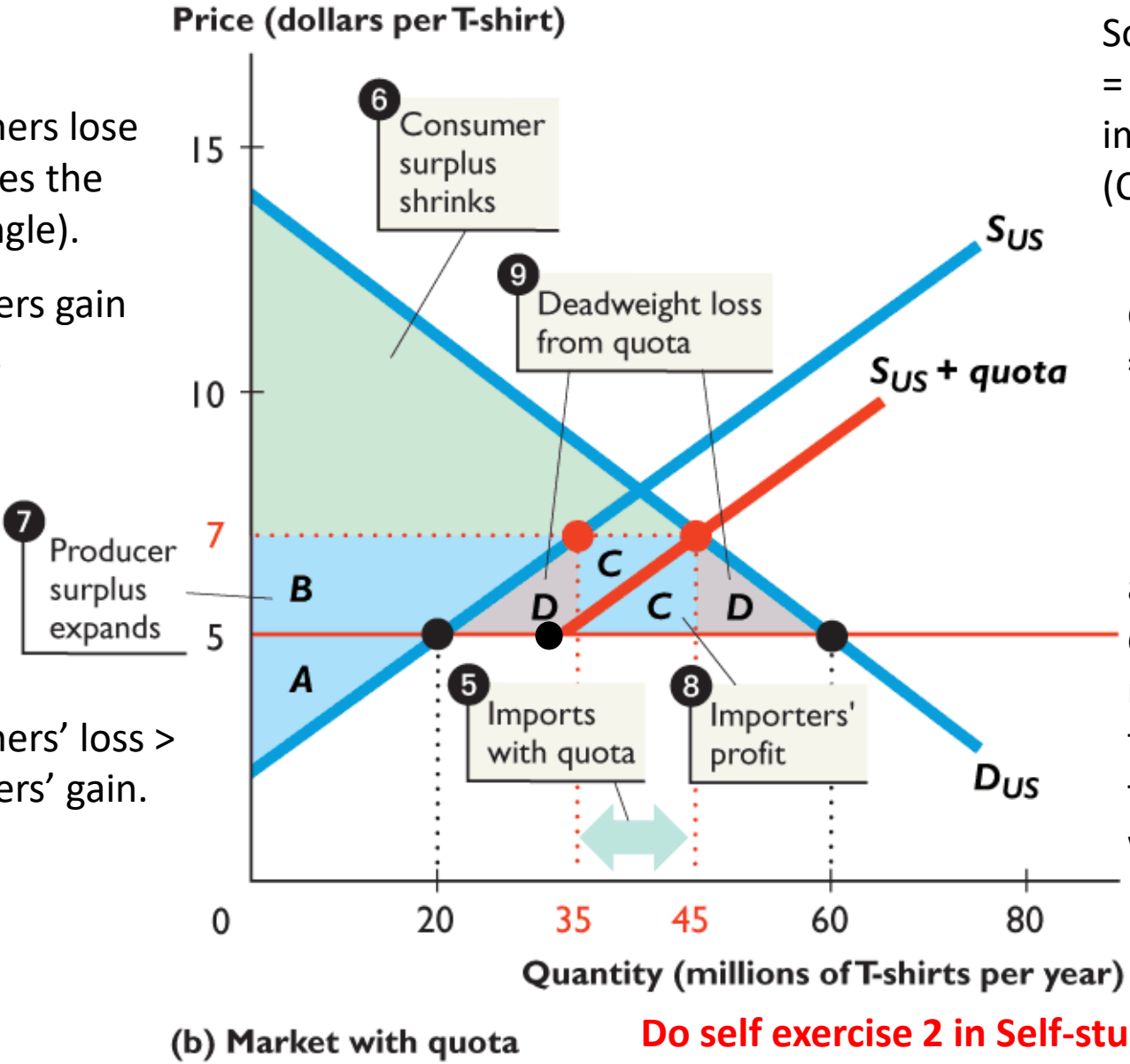
(b) Market with quota

Winners, Losers, and Social Loss from an Import Quota

US consumers lose (CS becomes the green triangle).

US producers gain (PS = A+B).

US consumers' loss > US producers' gain.



(b) Market with quota

Social surplus (SS) = CS + PS + importers' profit (C + C)

Reduction in SS due to quota = DWL = D + D

Importers' profit is also known as quota rent and may be earned by foreign firms. In this case, the DWL will be even larger.

Do self exercise 2 in Self-study Notes Unit 5 for self assessment.

SOMETHING TO THINK ABOUT

How does the Hong Kong government protect local domestic helpers by adopting Minimum Allowable Wage (MAW) for foreign domestic helpers (see Topic 2)?

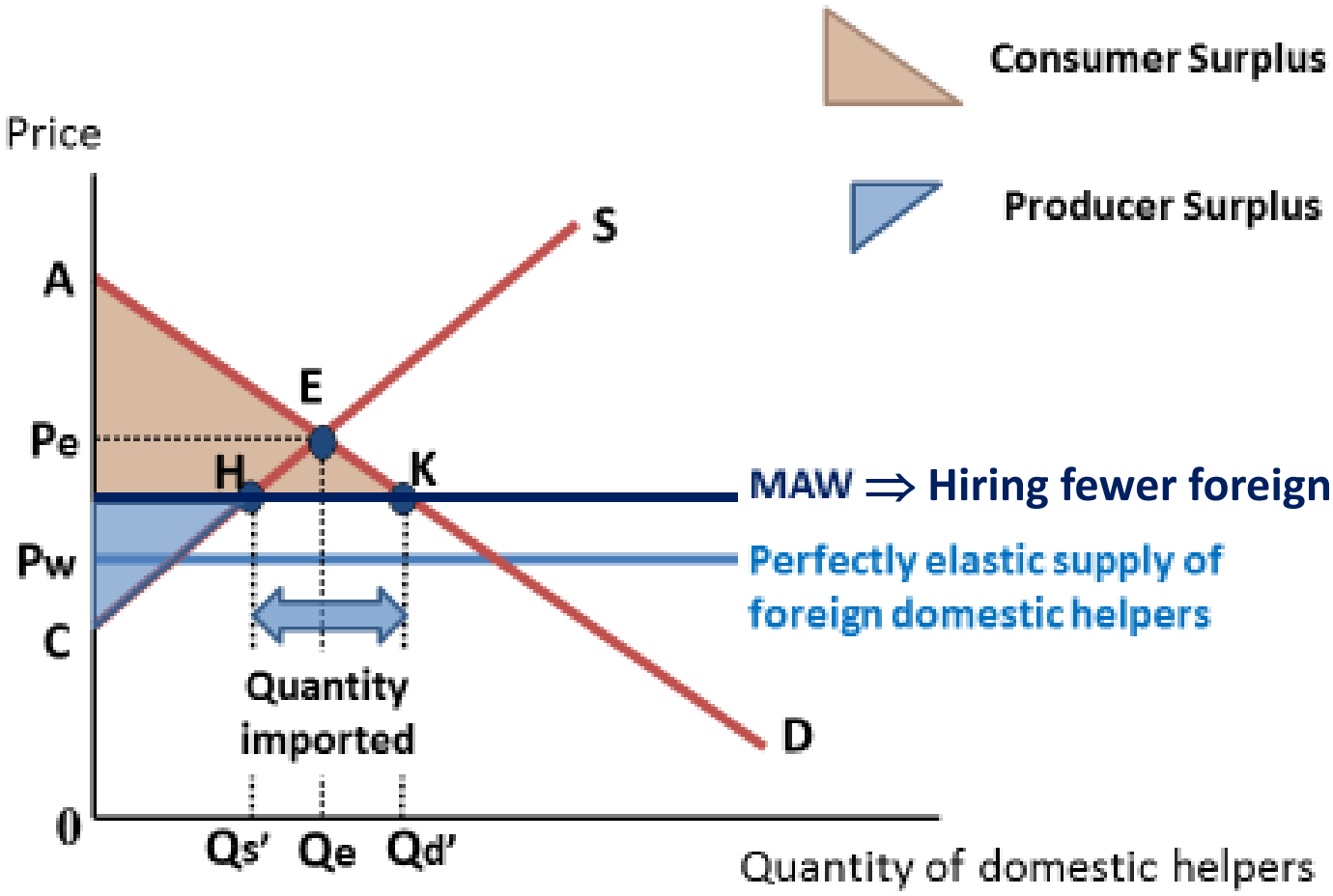
Whenever there is a shortage of workers in certain sector driving up local wage rate, the employers will ask the HK government to allow the importation of workers. Domestic helpers are one of the examples.

If the government allows the importation of domestic helpers at a wage rate determined in their home countries (usually less developed and poorer), what will happen?

They will replace a large number of local domestic helpers and drag down their wages!

Adopting MAW as an effective price floor (higher than their home-determined wage rates) will make them less attractive and protect local domestic helpers' job opportunities.

SOMETHING TO THINK ABOUT



Compare and contrast the effects of MAW & a tax rate (similar to tariff rate) on each imported foreign domestic helper.